Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Pre-Tutorial Assignment 1: Solutions

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1. Apply Ohm's Law, KCL and KVL to evaluate i_x and V_x from the circuit shown in Figure 1.

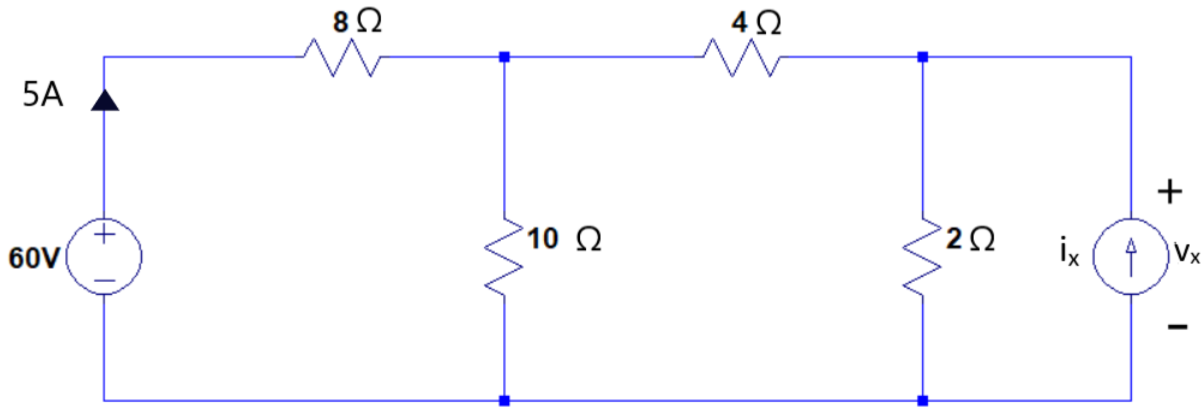


Figure 1: Evaluate i_x and v_x from the above circuit.

Solution

Figure 1 shows the different branch voltages and currents with their respective polarities and directions. The nodes are marked as A, B, C and D

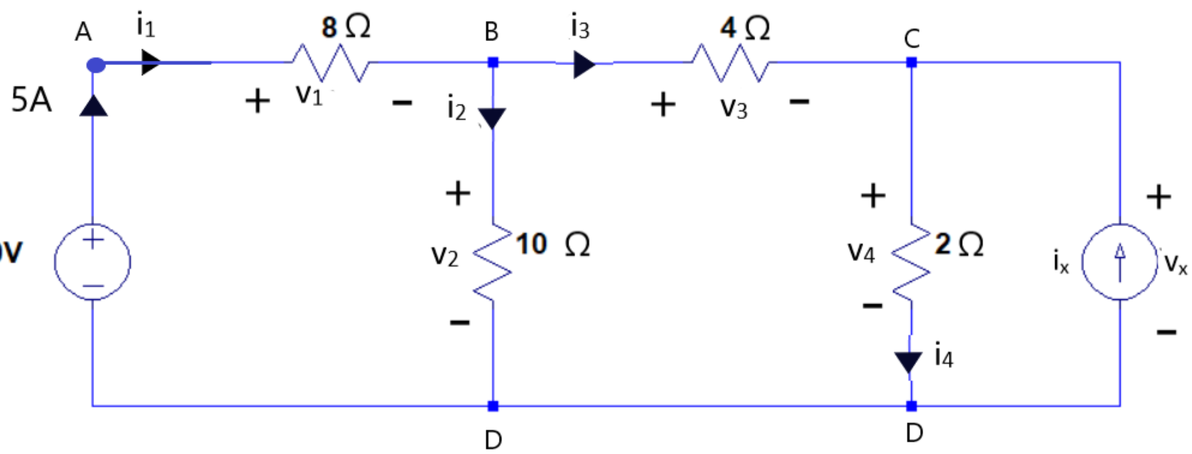


Figure 2: Branch voltages and currents are shown with their respective polarities and directions.

Apply KCL at node A: $i_1 \;=\; 5A$

$=> \; V_1 \;=\; 8\Omega \;\times\; 5A$

$=> V_1 = 40\,V$

Apply KVL : A-B-D-A (Voltage drop taken positive)

$V_1 + V_2 - 60 = 0$
$=> V_2 = 60 - V_1$
$=> V_2 = 20V$
$=> i_2 = \frac{20\,V}{10\,\Omega}$
$=> i_2 = 2A$

Apply KVL at Node B : $i_1 = i_2 + i_3$

$=> i_3 = 5 - 2$
$=> i_3 = 3A$
$=> V_3 = 3\,A * 4\,\Omega$
$=> V_3 = 12\,V$

Apply KVL : B-C-D-B (Voltage drop taken positive)

$V_3 + V_4 - V_2 = 0$
$=> V_4 = 20 - 12$
$=> V_4 = 8V$
$=> i_4 = \frac{8\,V}{2\,\Omega} = 4\,A$

Hence, $V_x = 8\,V$

Apply KCL at Node C : $i_3 + i_x = i_4$

$=> i_x = 4 - 3$

Hence, $i_x = 1\,A$

2. For each of the two circuits given below, compute the truth table for the output in terms of the inputs. Does the tables look familiar?

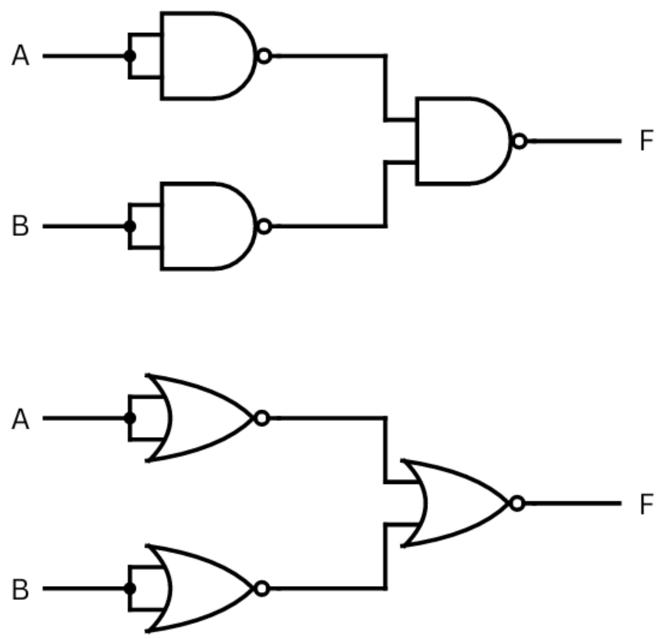
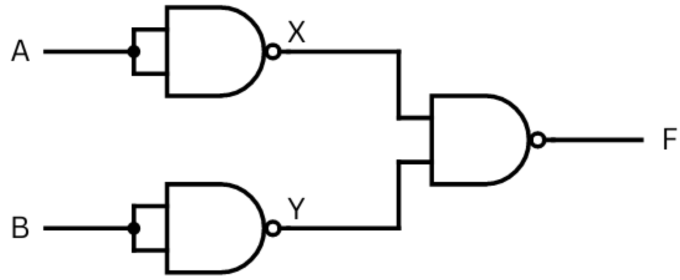


Figure 3: A logic gate circuit

Solution

Consider the variables X and Y shown below. Using the truth table of the 2-input NAND gate, we can compute what the values of X and Y are as a function of the inputs A and B.



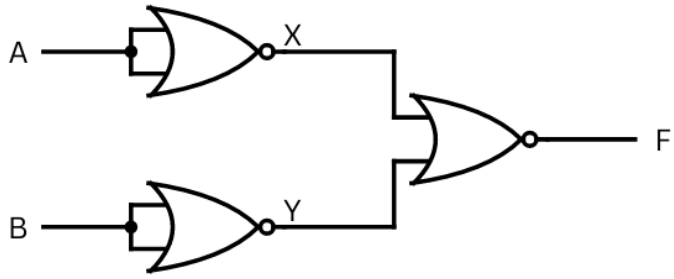
The truth table for F can be obtained as follows:

A	B	X	Y	F
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

The circuit can be recognized as the OR Gate from the truth table.

Consider the variables X and Y shown below. Using the truth table of the 2-input NOR gate, we can compute what the values of X and Y are as a function of the inputs A and B.

The truth table for F can be obtained as follows:



A	B	X	Y	F
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

The circuit can be recognized as the AND Gate from the truth table.